

Now, what two theorems must be proved to demonstrate the equivalence of definitions 1 and 2? Can you formulate these theorems?

Reader: We have to prove two theorems, one direct and the other converse. We want to prove that definition 2 follows from definition 1 and vice-versa (i.e. definition 1 follows from definition 2).

Author: Correct. First, I shall prove the following

Theorem: If a number b is the limit of a function $f(x)$ at a point a in terms of definition 1, it is the limit of the function $f(x)$ at a in terms of definition 2 as well.

Proof: Since b is the limit of the function $f(x)$ at point a in terms of definition 1 (this is given), consequently, for any $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - b| < \varepsilon$ for all $x \neq a$ from a δ -neighbourhood of point a . Then we "construct" an arbitrary sequence (x_n) , requiring that it be convergent to point a (any x_n belong to the domain of the function and $x_n \neq a$ for any n). As a result we obtain a sequence of the corresponding values of the function (the sequence $[f(x_n)]$). We want to prove that the sequence $[f(x_n)]$ is convergent to b .

First, I select an arbitrary $\varepsilon > 0$. I should find a number N such that $|f(x_n) - b| < \varepsilon$ for all $n > N$.

I cannot immediately find such N for an arbitrary ε . However, I can indicate for an arbitrary ε such δ that $|f(x) - b| < \varepsilon$ if $|x - a| < \delta$. Then I take this δ and find a sequence (x_n) convergent to a . Evidently, since (x_n) is convergent to a , δ (as any other positive real number) can be placed in correspondence with a number N such that $|x_n - a| < \delta$ for all $n > N$. And, consequently, we also have that $|f(x_n) - b| < \varepsilon$ for all $n > N$. Hence we find that the thus found number N is actually the desired number. It proves the convergence of the sequence $[f(x_n)]$ to b . Since the sequence (x_n) , which is convergent to a , was chosen ("constructed") arbitrarily, we conclude that the theorem's proof is completed.

If the line of reasoning is clear to you, try briefly to recapitulate the logical structure of the proof.